

What exactly existing companies are doing related to A.E.G.I.S?

Recall (Official Microsoft)

<https://support.microsoft.com/en-US/Windows/privacy/privacy-and-control-over-your-recall-experience>

- Recall (preview) is an experience exclusive to Copilot+ PCs that will help you easily find and remember things you've seen using natural language. To help provide you with that 'photographic' memory, you can opt in to saving snapshots of your screen periodically. You can quickly search your snapshots to find things on your Copilot+ PC. For example, you can search for content you've seen in apps, websites, images, and documents.
- Recall processes your content locally on the Copilot+ PC and securely stores it on your device.
- You can filter out apps and websites from being saved as snapshots.
- Snapshots and associated data are stored locally on the device. Recall does not share snapshots or associated data with Microsoft or third parties

What matches with ages?

Microsoft Recall closely matches the **Context Retrieval Module** and **Memory Layer** of AEGIS. Like AEGIS, Recall helps users retrieve previously viewed information by understanding natural language queries and maintaining a searchable history of user activity. It captures screen snapshots, allows users to search content from applications, websites, images, and documents, and stores all information locally on the device with privacy controls. This aligns with AEGIS's goal of maintaining contextual awareness and preserving user privacy through local memory management. However, Recall mainly focuses on remembering and retrieving past information, whereas AEGIS extends this concept by using the retrieved context for intelligent task planning and execution.

Anthropic – Claude Computer Use

<https://platform.claude.com/docs/en/agents-and-tools/tool-use/computer-use-tool>

- Computer use is a beta feature that enables Claude to interact with desktop environments
- Screenshot capture: See what's currently displayed on screen
- Mouse control: Click, drag, and move the cursor
- Keyboard input: Type text and use keyboard shortcuts.
- Desktop automation: Interact with any application or interface.

What matches with ages?

Claude Computer Use strongly matches the Action Execution Layer of AEGIS. It enables an AI agent to interact directly with desktop environments by viewing screenshots, controlling the mouse, typing through the keyboard, and operating desktop applications. These capabilities are similar to AEGIS's objective of executing user tasks across different desktop applications. However, Claude primarily focuses on executing interface actions, while AEGIS combines execution with context retrieval, memory, planning, and decision-making to complete complex user workflows.

OpenAI

<https://developers.openai.com/api/docs/guides/tools-computer-use>

- Computer use lets a model operate software through the user interface
- It can inspect screenshots, return interface actions for your code to execute
- The model looks at the current UI through a screenshot, returns actions such as clicks, typing, or scrolling, and your harness executes those actions in a browser or computer environment.
- After the actions run, your harness sends back a new screenshot so the model can see what changed and decide what to do next
- This makes the built-in path intuitive for tasks that a person could complete through a UI, such as navigating a site, filling out a form, or stepping through a multistage workflow

- Send a task to the model with the computer tool enabled.
- Inspect the returned `computer_call`.
- Run every action in the returned `actions[]` array, in order.
- Capture the updated screen and send it back as `computer_call_output`.
- Repeat until the model stops returning `computer_call`.

What matches with ages?

OpenAI's Computer Use feature closely aligns with the Task Planning Agent, Action Execution Layer, and continuous execution loop proposed in AEGIS. The model observes the user interface through screenshots, plans the next action, performs interface operations such as clicking, typing, and scrolling, and continuously receives updated screenshots to determine subsequent actions until the task is complete. This workflow is very similar to the planning-execution-feedback cycle described in AEGIS. However, AEGIS additionally integrates long-term memory, contextual understanding, and personalized desktop knowledge to improve future task execution

Apple Intelligence

https://developer.apple.com/documentation/technologyoverviews/apple-intelligence/?utm_source=chatgpt.com

- Apple Intelligence is the personal intelligence system behind many built-in capabilities, and you use it to create contextually relevant and personal experiences for people
- Integrate your app's actions and data with Apple Intelligence to enhance system features like Siri, Spotlight, and Shortcuts.
- Teach the system about your app's actions and data
- An app intent is a custom type that encapsulates one of your app's actions
- App entities represent the data you need to perform those actions
- If someone performs the same action every day, the system might preemptively suggest that action at the appropriate time
- Apple Intelligence uses donated app entities to identify data in your app's interface, which Siri can use as additional context during a conversation.

What matches with ages?

Apple Intelligence mainly corresponds to the Intent Understanding Module, Context Retrieval Module, and personalized user assistance components of AEGIS. It understands application actions, learns from user behavior, provides proactive suggestions, and uses application data to improve Siri and system intelligence. Similar to AEGIS, it utilizes contextual information and previous user interactions to provide more relevant assistance. However, Apple Intelligence is primarily designed for Apple's ecosystem, whereas AEGIS aims to function as a desktop copilot capable of planning and executing multi-step workflows across the operating system

Google Gemini Computer Use

https://ai.google.dev/gemini-api/docs/computer-use?utm_source=chatgpt.com

- The Computer Use tool lets you build browser, mobile, and desktop control agents that interact with and automate tasks
- Using screenshots, the model can 'see' a computer screen, and 'act' by generating specific UI actions like mouse clicks and keyboard inputs
- With Computer Use, you can build agents that:
 1. Automate repetitive data entry or form filling on websites.
 2. Perform automated testing of web applications and user flows.
 3. Conduct research across various websites.
- To build an agent with the Computer Use model, you need to set up a continuous loop between your application and the API

1. Send a request to the model
2. Receive the model response
3. Execute the received action
4. Capture the new environment state
5. This process then repeats until the task is completed.
 - The response may also include a safety_decision... regular/allowed, require_confirmation, or blocked
 - Run your agent in a sandboxed VM or container to isolate it from your host system.

What matches with ages?

Google Gemini Computer Use closely matches the Action Execution Layer, Task Planning Module, and Safety Layer of AEGIS. It allows AI agents to automate browser, desktop, and mobile tasks by observing screenshots, generating user interface actions, and repeatedly interacting with the environment until the assigned task is completed. It also includes a safety decision mechanism that can allow, confirm, or block potentially sensitive actions before execution. These concepts closely resemble AEGIS's planning, execution, environment monitoring, and secure task automation architecture.

Lang chain

<https://docs.langchain.com/build-overview>

- LangChain is an open source framework with a prebuilt agent architecture and integrations for any model or tool—so you can build agents that adapt as fast as the ecosystem evolves.
- Build completely custom agents and applications powered by LLM
- LangChain's agents are built on top of LangGraph. This allows us to take advantage of LangGraph's durable execution, human-in-the-loop support, persistence, and more.
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- LangChain standardizes how you interact with models so that you can seamlessly swap providers and avoid lock-in

What matches with ages?

LangChain is different from the previous companies because it is not an end-user product but an agent development framework. It closely matches the internal architecture proposed for AEGIS rather than its functionality. LangChain provides reusable components for building intelligent agents with tool usage, durable execution, human-in-the-loop support, persistence, and integration with different language models. These concepts are highly relevant to AEGIS because its Task Planning Agent, Memory Layer, Tool Integration, and execution workflow can all be implemented using an architecture similar to LangChain. Rather than competing with AEGIS, LangChain can serve as one of the technologies used to build it.

What we can different from existing contributions?

1. Learn User Habits

- AEGIS can observe the user's daily routine and learn frequently performed tasks.
- It can identify repeated workflows, such as opening the same applications every morning.
- Based on these habits, AEGIS can suggest or automate routine tasks.
- This makes the assistant more personalized over time.

2. Explain Before Taking Any Action

- Before performing any task, AEGIS should explain what actions it is going to perform.

- It should ask for the user's approval before executing important actions.
- This increases transparency and gives users full control over the system.

3. Student-Focused Assistant

- Unlike existing assistants designed for general users, AEGIS can be specially designed for students.
- It can help manage lecture notes, assignments, research papers, and study materials.
- It can support study planning and academic productivity.

4. Personal Knowledge Base

- AEGIS can organize the user's documents, notes, PDFs, and project files.
- It can search and retrieve information from personal files quickly.
- This creates a personalized knowledge base for the user.

5. Multi-Application Workflow Automation

- AEGIS can complete a task by working across multiple desktop applications.
- For example, it can open a browser, access an email account, attach a document, and prepare an email in one workflow.
- This reduces the need for manual switching between applications.

6. Productivity Dashboard

- AEGIS can track how users spend their time on different applications.
- It can generate reports showing productive hours, application usage, and completed tasks.
- It can also provide suggestions to improve productivity.

7. Smart File Organization

- AEGIS can automatically organize files into appropriate folders.
- It can detect duplicate or old files and recommend cleaning them.
- This helps users keep their system organized.

8. Workflow Suggestions

- AEGIS can observe frequently repeated task sequences.
- It can suggest automating those workflows.
- This saves time and improves efficiency.

9. Task History and Recovery

- AEGIS can maintain a history of completed tasks.
- If the computer shuts down unexpectedly, it can help restore the previous working session.
- This improves continuity and reduces lost work.

10. Privacy and Permission Control

- AEGIS can ask for user confirmation before accessing sensitive files or performing important actions.
- Users remain in control of their personal data.
- This improves trust and system security.

11. Context-Aware Assistance

- AEGIS can understand what the user is currently doing.
- It can provide suggestions based on the current application, opened documents, or ongoing task.
- This makes assistance more relevant and useful.

12. Personalized Recommendations

- AEGIS can recommend useful actions based on previous user behavior.
- For example, it can suggest opening frequently used applications or documents at the right time.
- These recommendations become more accurate as the system learns user preferences.